

Commissioning ZF Autotroll Inside

Electronic Equipment
BlueVision® | NewGeneration

E-INB-M-060-2018/00E



Power. Passion. Partnership.

Inhaltsverzeichnis

1 SAFETY	3
1.1 GENERAL REQUIREMENTS	3
1.2 PERSONNEL AND ORGANIZATIONAL REQUIREMENTS	4
1.3 SAFETY REQUIREMENTS FOR COMMISSIONING AND OPERATION	5
1.4 CONVENTIONS FOR WARNINGS USED IN THE TEXT	6
2 CONFIGURATION / BVNG CONFIGURATOR	7
2.1 RCS-TAB	7
2.1.1 Select trolling type "ZF Autotroll Inside"	7
2.1.2 Select parameter set for the autotroll control	7
2.2 GEAR-TAB	8
2.2.1 Gear-Tab: Common	8
2.2.1.1 Enter / check transmission ratio	8
2.2.2 Gear-Tab: Sensor Selection 1	8
2.2.2.1 Modify gear settings	8
2.2.2.2 Select shaft speed sensor and engine speed sensor	9
2.2.3 Gear-Tab: Shaft Speed	10
2.2.3.1 Check the number of teeth of the shaft speed sensor	10
2.2.4 Gear-Tab: Engine Speed	10
2.2.4.1 Check the setting of the engine speed sensor	10
3 COMMISSIONING	11
4 DIAGNOSIS	12
4.1 MONITORING DURING AUTOMATIC COMMISSIONING PROCEDURE	13
4.2 CHECK SHAFT SPEED	14
4.3 CHECK ENGINE SPEED	14
4.4 ERROR HANDLING / ALARMING	15
5 SENSOR WIRING (SYSTEM CABLE –W623.4).....	16
6 NUMBER OF TEETH (IMPULSES PER ROTATION) OUTPUT SHAFT (SHAFT SPEED).....	17

1 Safety

1.1 General requirements

General information

This document specifies recommended settings for adjusting the responses of the BlueVision | NewGeneration automation system when a crash stop is initiated (SPU software version Var1_Ed01_040 and higher). The document is not intended as an alternative to formal training in initial start-up procedures and servicing at the Training Center.

The shipyard must be made aware of the absolute necessity of conducting a crash stop maneuver during trials in order to verify that the necessary settings have been made to ensure that the vessel responds safely and reliably in the event of an emergency.

Make a written entry in the commissioning record stating reasons, and have the shipyard confirm accordingly, if it is not possible to configure the crash stop function in the course of sea trials, or should the shipyard or customer fail to facilitate this.

Observe all generally applicable, national, statutory and otherwise relevant accident prevention and environmental protection regulations in addition to the instructions provided in this publication. This engine has been manufactured according to the current state of the art and applicable specifications and regulations. Nevertheless, the engine may endanger personnel or incur damage if:

- It is not put to its intended use
- It is not operated, maintained and repaired by trained personnel
- It is changed or modified
- Safety notes are disregarded

Intended use

The engine is intended for use in accordance with contractual agreements and/or the purpose for which it was designated on delivery. Any other use above or beyond this shall be deemed improper. The engine manufacturer accepts no liability whatsoever for damage incurred by improper use.

The user shall bear all risk in such case.

Intended use includes compliance with the operating instructions and strict observance of maintenance and repair procedures.

Changes or modifications

Unauthorized changes to the engine compromise safety.

MTU accepts no liability whatsoever for damage resulting from unauthorized changes or modifications and all warranties will be rendered null and void.

Spare parts

Use genuine MTU spare parts only when replacing components or assemblies. MTU accepts no liability whatsoever for damage resulting from the use of non-MTU spare parts and all warranties will be rendered null and void.

Corrective work on components

Work shall be entrusted to an authorized MTU workshop in case of repair or engine overhaul.

1.2 Personnel and organizational requirements

Personnel requirements

Work on the product shall be performed by trained and qualified personnel only:

- Trained at the manufacturer's Training Center
- Qualified in machinery and plant engineering

The user shall define personnel responsible for operation, maintenance, repair and transportation.

Organizational measures

This publication shall be made available to all personnel involved in operation, maintenance, repair or transportation.

It must be kept ready to hand in the vicinity of the engine and be readily available to operating, maintenance, repair and transportation personnel at all times.

This publication serves as a basis for instructing personnel in matters of handling and repair of the engine. Safety instructions, in particular, must be read and clearly understood.

This applies especially to personnel only working occasionally on the engine. Such personnel must receive repeated instruction to refresh their knowledge.

Use the Spare Parts Catalog to identify spare parts for maintenance and repair work.

Workwear and protective equipment

Wear safety clothing at all times when working on or around the engine.

Wear suitable personal safety equipment appropriate to the type of work concerned.

1.3 Safety requirements for commissioning and operation

Safety requirements for commissioning

The assembly/system must have been properly installed and accepted in accordance with MTU specifications before being put into operation for the first time.

Before starting up any device or system, ensure that:

- All maintenance and repair work has been completed.
- All loose parts have been removed from rotating parts of the machine.
- All personnel is clear of the danger zone around moving parts of the machine.

Immediately after starting up the device or the system ensure that all controls and display instruments including monitoring, signaling and alarm systems are operating correctly.

Safety requirements for operation

Emergency procedures must be practiced on a regular basis.

Operators must be familiar with the controls and displays.

Operators must be aware of the effects of any action they perform.

During operation, the indicators, displays and monitoring assemblies must be supervised with respect to current operating states, violation of limit values as well as warnings and alarm messages.

If a fault in the system is detected or signaled by the system:

- Inform the person in charge.
- Evaluate the signal or message.
- Take any necessary counteraction, e.g. emergency engine stop.

Engine operation

The following requirements must be fulfilled prior to engine starting:

- Wear ear protectors.
- Ensure that the engine room is well-ventilated.
- Do not inhale exhaust gases emitted by the engine.
- Check exhaust pipework for leakage and ensure that exhaust gases are vented to atmosphere.
- Wipe up any leaked or spilled fluids and lubricants or use a suitable binding agent.
- Protect battery and alternator terminals or cabling from inadvertent contact.
- Never undo water, oil, fuel, compressed air or hydraulic lines when the engine is running.

Electrical equipment

Some of the components are live when electrical equipment is in operation.

Observe warnings and instructions which apply to such equipment.

1.4 Conventions for warnings used in the text

DANGER 	In case of imminent threat or danger. Consequences: Death, severe or irreversible injury! • Corrective action.
WARNING 	In case of possible threat or danger. Consequences: Death, severe or irreversible injury! • Corrective action.
CAUTION 	In case of possible threat or danger. Consequences: Minor injury! • Corrective action.
ATTENTION 	In case of possible damage to the product. Consequences: Material damage! • Corrective action. • Additional information about the product.

Warnings

Personnel charged with operation, maintenance, repair or transportation must have read and understood this manual including all the safety instructions and warnings provided therein.

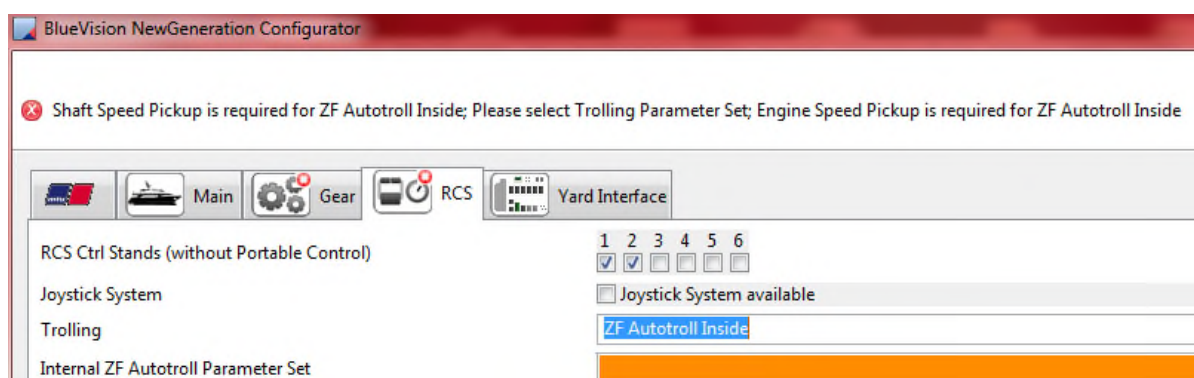
2 Configuration / BVNG Configurator

The MTU system software is delivered preconfigured, so that the settings regarding "ZF Autotroll Inside" are made by MTU.

The settings listed below are used for retrofitting and thus for the configuration to be carried out on site or to check the settings during commissioning.

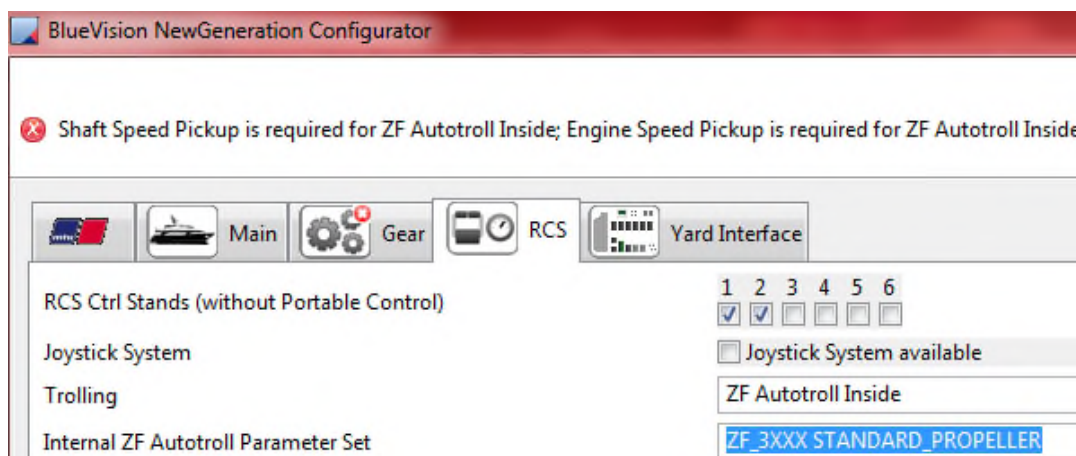
2.1 RCS-Tab

2.1.1 Select trolling type "ZF Autotroll Inside"



2.1.2 Select parameter set for the autotroll control

There is a separate parameter set for each gear unit family.
In addition, a distinction is made in surface propellers and standard FPP.

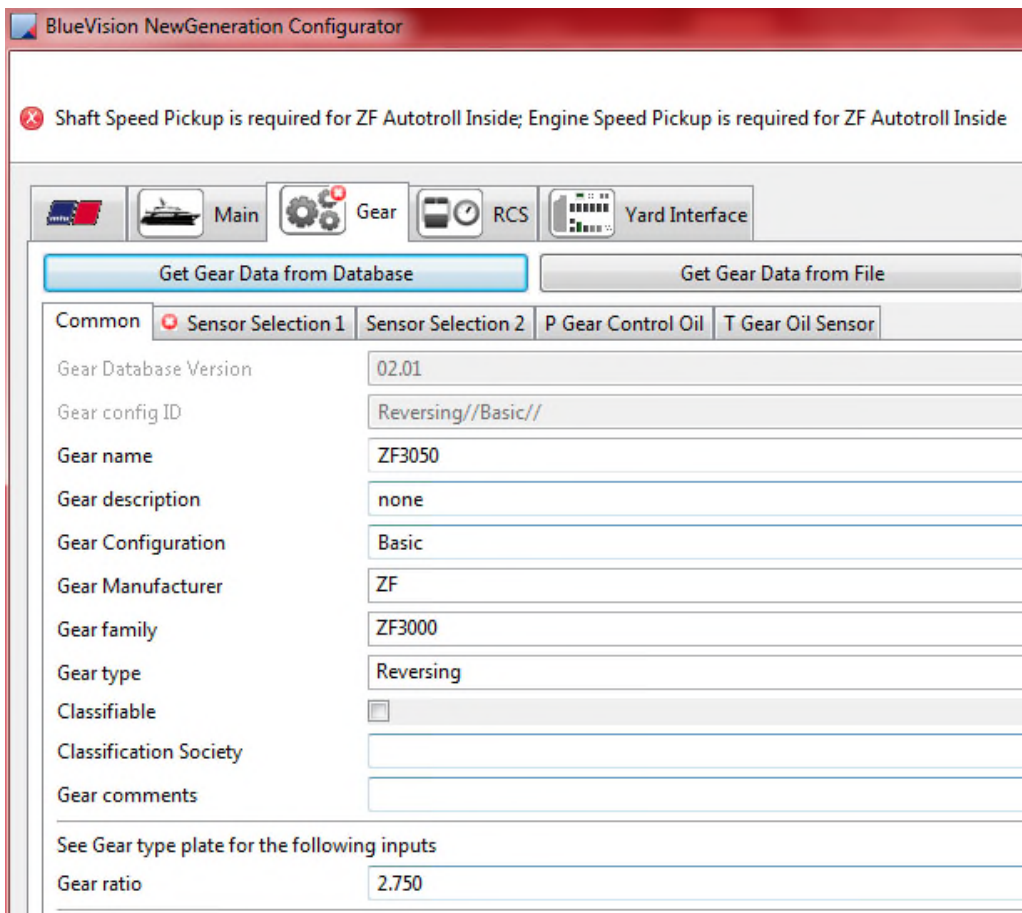


2.2 Gear-Tab

2.2.1 Gear-Tab: Common

2.2.1.1 Enter / check transmission ratio

Check whether the set gear ratio is correct, otherwise correct.
This is essential for the later calculation.



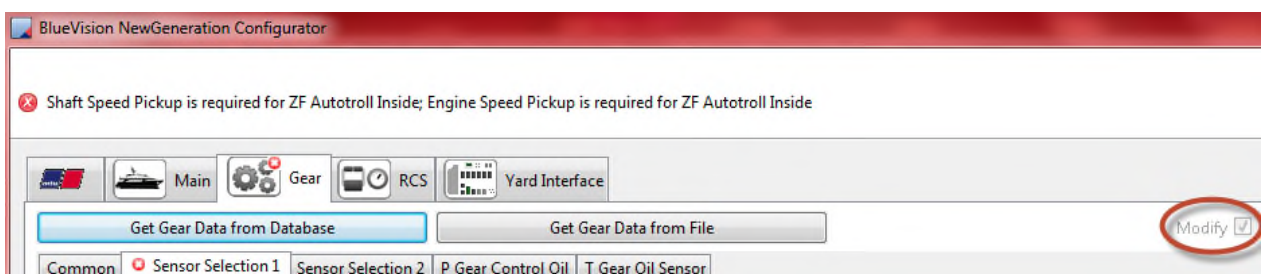
The screenshot shows the 'BlueVision NewGeneration Configurator' window with the 'Gear' tab selected. A warning message at the top states: 'Shaft Speed Pickup is required for ZF Autotroll Inside; Engine Speed Pickup is required for ZF Autotroll Inside'. The 'Gear' tab is active, and the 'Common' sub-tab is selected. The 'Get Gear Data from Database' button is highlighted. The form contains the following fields:

Common	Sensor Selection 1	Sensor Selection 2	P Gear Control Oil	T Gear Oil Sensor
Gear Database Version	02.01			
Gear config ID	Reversing//Basic//			
Gear name	ZF3050			
Gear description	none			
Gear Configuration	Basic			
Gear Manufacturer	ZF			
Gear family	ZF3000			
Gear type	Reversing			
Classifiable	☐			
Classification Society				
Gear comments				
See Gear type plate for the following inputs				
Gear ratio	2.750			

2.2.2 Gear-Tab: Sensor Selection 1

2.2.2.1 Modify gear settings

Make gear data settings modifiable via "Modify".



This screenshot is identical to the previous one, but with a red circle highlighting the 'Modify' button in the bottom right corner of the window.

2.2.2.2 Select shaft speed sensor and engine speed sensor

Select „Shaft Speed Pickup 1 Available” and “Engine Speed Pickup Available”.

BlueVision NewGeneration Configurator

✖ Shaft Speed Pickup is required for ZF Autotroll Inside; Engine Speed Pickup is required for ZF Autotroll Inside

Main Gear RCS Yard Interface

Get Gear Data from Database Get Gear Data from File

Common Sensor Selection 1 Sensor Selection 2 P Gear Control Oil T Gear Oil Sensor

P Gear Control Oil Sensor Available	☒
P Gear Control Oil Redundant Sensor Available	☐
P Gear Control Oil Redundant Switch Type	None
P Gear Lube Oil Sensor Available	☐
P Gear Lube Oil Redundant Sensor Available	☐
P Gear Lube Oil Redundant Switch Type	None
T Gear Oil Sensor Type	PT1000
T Gear Thrust Bearing Sensor Type	None
T Gear Oil Aftercooler Sensor Type	None
Gear Oil Filter Clogged Sensor Type	None
Gear Oil Level Sensor Type	None
Shaft Speed Pickup 1 Available	☐
Engine Speed Pickup Available	☐
Engine Speed Pickup Index	2

BlueVision NewGeneration Configurator

Main Gear RCS Yard Interface

Get Gear Data from Database Get Gear Data from File

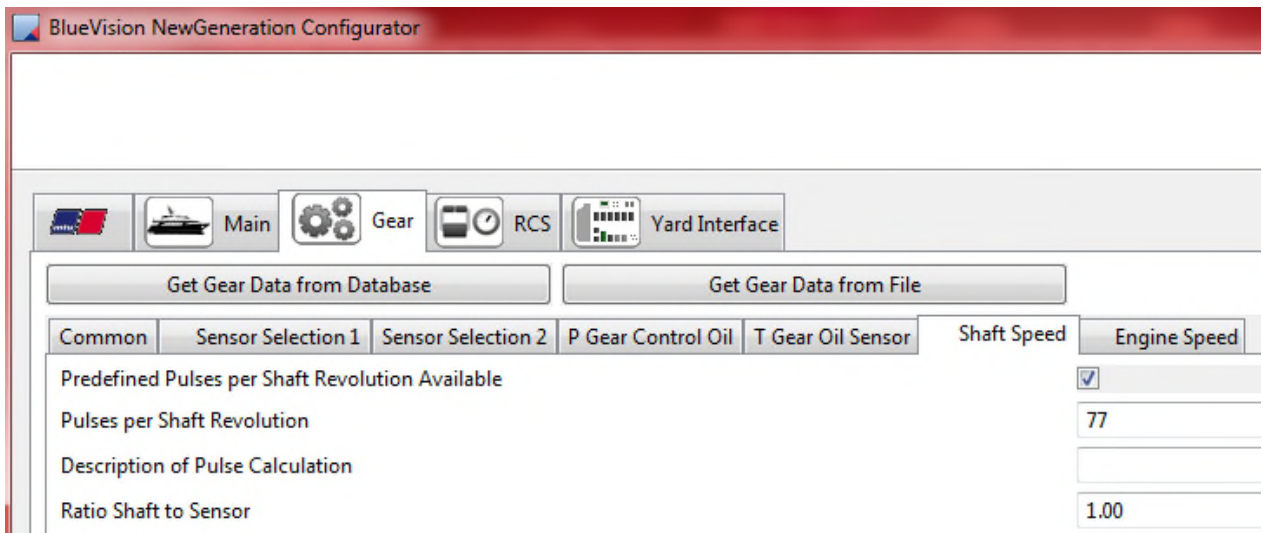
Common Sensor Selection 1 Sensor Selection 2 P Gear Control Oil T Gear Oil Sensor Shaft Speed Engine Speed

P Gear Control Oil Sensor Available	☒
P Gear Control Oil Redundant Sensor Available	☐
P Gear Control Oil Redundant Switch Type	None
P Gear Lube Oil Sensor Available	☐
P Gear Lube Oil Redundant Sensor Available	☐
P Gear Lube Oil Redundant Switch Type	None
T Gear Oil Sensor Type	PT1000
T Gear Thrust Bearing Sensor Type	None
T Gear Oil Aftercooler Sensor Type	None
Gear Oil Filter Clogged Sensor Type	None
Gear Oil Level Sensor Type	None
Shaft Speed Pickup 1 Available	☒
Shaft Speed Pickup 2 Available	☐
Engine Speed Pickup Available	☒
Engine Speed Pickup Index	2

2.2.3 Gear-Tab: Shaft Speed

2.2.3.1 Check the number of teeth of the shaft speed sensor

Check the number of teeth (impulses per rotation) of the shaft speed sensor (see [chapter 6](#)) and correct if necessary, if this should be different from the presetting of the database.



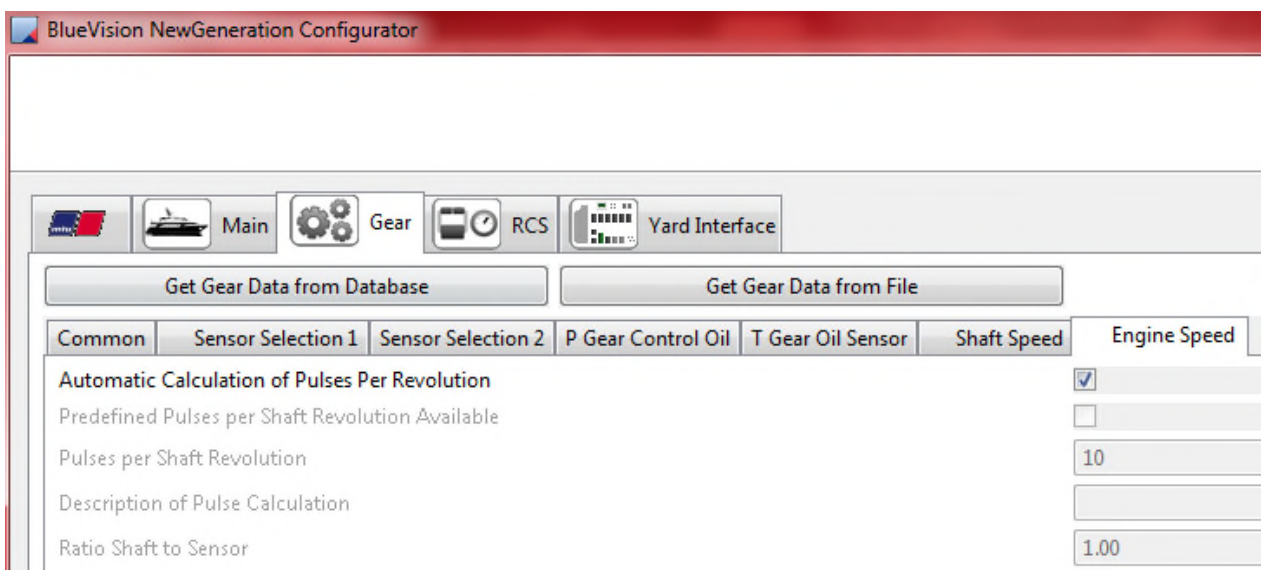
The screenshot shows the 'BlueVision NewGeneration Configurator' window. The 'Gear' tab is selected, and the 'Shaft Speed' sub-tab is active. The 'Get Gear Data from Database' button is highlighted. The 'Predefined Pulses per Shaft Revolution Available' checkbox is checked. The 'Pulses per Shaft Revolution' field is set to 77. The 'Description of Pulse Calculation' field is empty. The 'Ratio Shaft to Sensor' field is set to 1.00.

Common	Sensor Selection 1	Sensor Selection 2	P Gear Control Oil	T Gear Oil Sensor	Shaft Speed	Engine Speed
Predefined Pulses per Shaft Revolution Available						☒
Pulses per Shaft Revolution						77
Description of Pulse Calculation						
Ratio Shaft to Sensor						1.00

2.2.4 Gear-Tab: Engine Speed

2.2.4.1 Check the setting of the engine speed sensor

Check that the engine speed sensor setting is set to "Automatic Calculation of Pulses Per Revolution". Correct if necessary.



The screenshot shows the 'BlueVision NewGeneration Configurator' window. The 'Gear' tab is selected, and the 'Engine Speed' sub-tab is active. The 'Get Gear Data from Database' button is highlighted. The 'Automatic Calculation of Pulses Per Revolution' checkbox is checked. The 'Predefined Pulses per Shaft Revolution Available' checkbox is unchecked. The 'Pulses per Shaft Revolution' field is set to 10. The 'Description of Pulse Calculation' field is empty. The 'Ratio Shaft to Sensor' field is set to 1.00.

Common	Sensor Selection 1	Sensor Selection 2	P Gear Control Oil	T Gear Oil Sensor	Shaft Speed	Engine Speed
Automatic Calculation of Pulses Per Revolution						☒
Predefined Pulses per Shaft Revolution Available						☐
Pulses per Shaft Revolution						10
Description of Pulse Calculation						
Ratio Shaft to Sensor						1.00

3 Commissioning

Trolling function is not available after the first start of the system and can't be selected/activated.
If you try to select a trolling mode (SCRS or TRL) the display shows the operating instruction
"System Function Limited."

In order to make the trolling function available, the automatic commissioning procedure must first be carried out during commissioning, which calculates the number of teeth on the drive side (engine speed sensor), checks all values for plausibility and finally releases trolling.

The automatic commissioning procedure, in which the engine must be running and the gear must be engaged, is completed after about 30-60s.

The following conditions (per shaft) must be met in order for the automatic commissioning procedure to be started:

- ☒ Engine running
- ☒ Gearbox engaged
- ☒ EngineSpeed > 0 rpm
- ☒ Control lever is in Cruising mode (CRS-button).
This means Trolling not activ (Cmd_TrollingOn and Autotroll_Soll_Enable_Ventil)
- ☒ **pZF_AT_CommissioningDone** is set to „FALSE“ (calculation not yet carried out)
- ☒ Trolling type „ZF Autotroll Inside“ ist configured (see BVNG configurator, [chapter 2.1.1](#))

DANGER 	Rotating and moving engine parts. Risk of crushing, danger of parts of the body being caught or pulled in! • Before barring the engine, make sure that there are no persons in the engine's danger zone. • After finishing work on the engine, make sure that all safety devices are put back in place and all tools are removed from the engine.
--	---

WARNING 	Uncontrolled vessel movements due to sudden control lever movement. Danger of injury, major material damage! • Actuate control lever with caution.
---	--

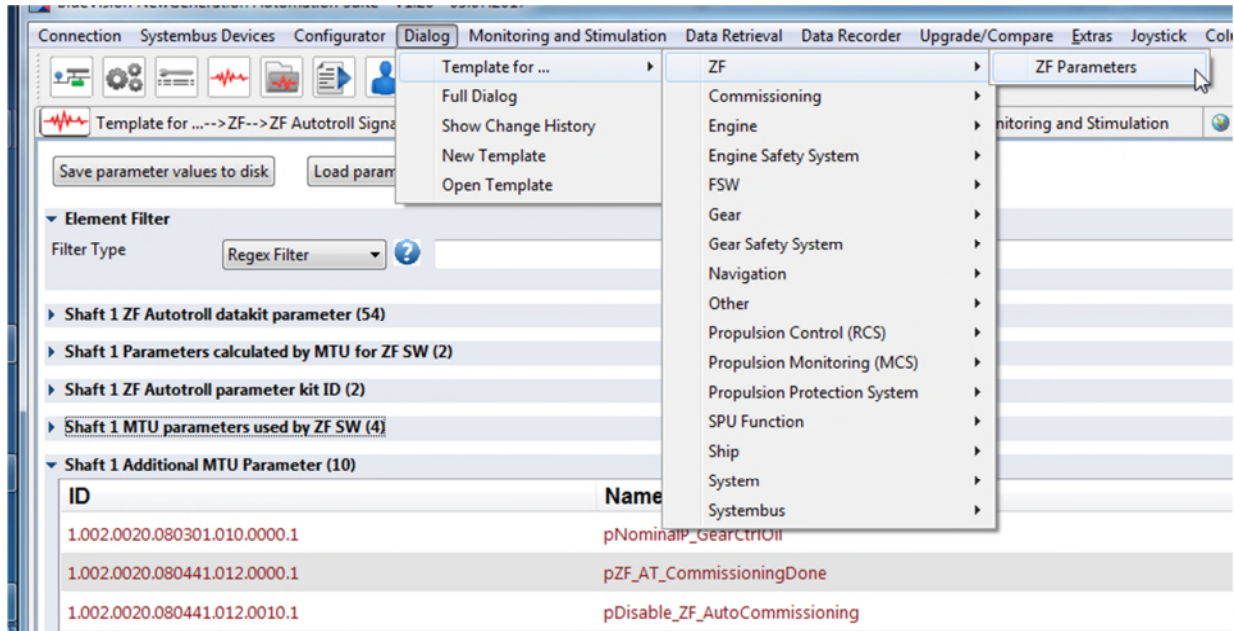
Procedure (should be carried out on all shafts at the same time):

- ☒ Turn on the system ready for operation
- ☒ Start the engines
- ☒ Engage gearboxes and keep engine speed at a constant level
(e.g. Detent Ahead Position at control lever)
- ☒ Maintain this state for at least 30s to about 60s
(Duration of the automatic commissioning procedure / The time during maneuvering is not enough to complete the procedure)
- ☒ Now Trolling mode respectively Smart Cruise mode is ready to be activated at the control lever

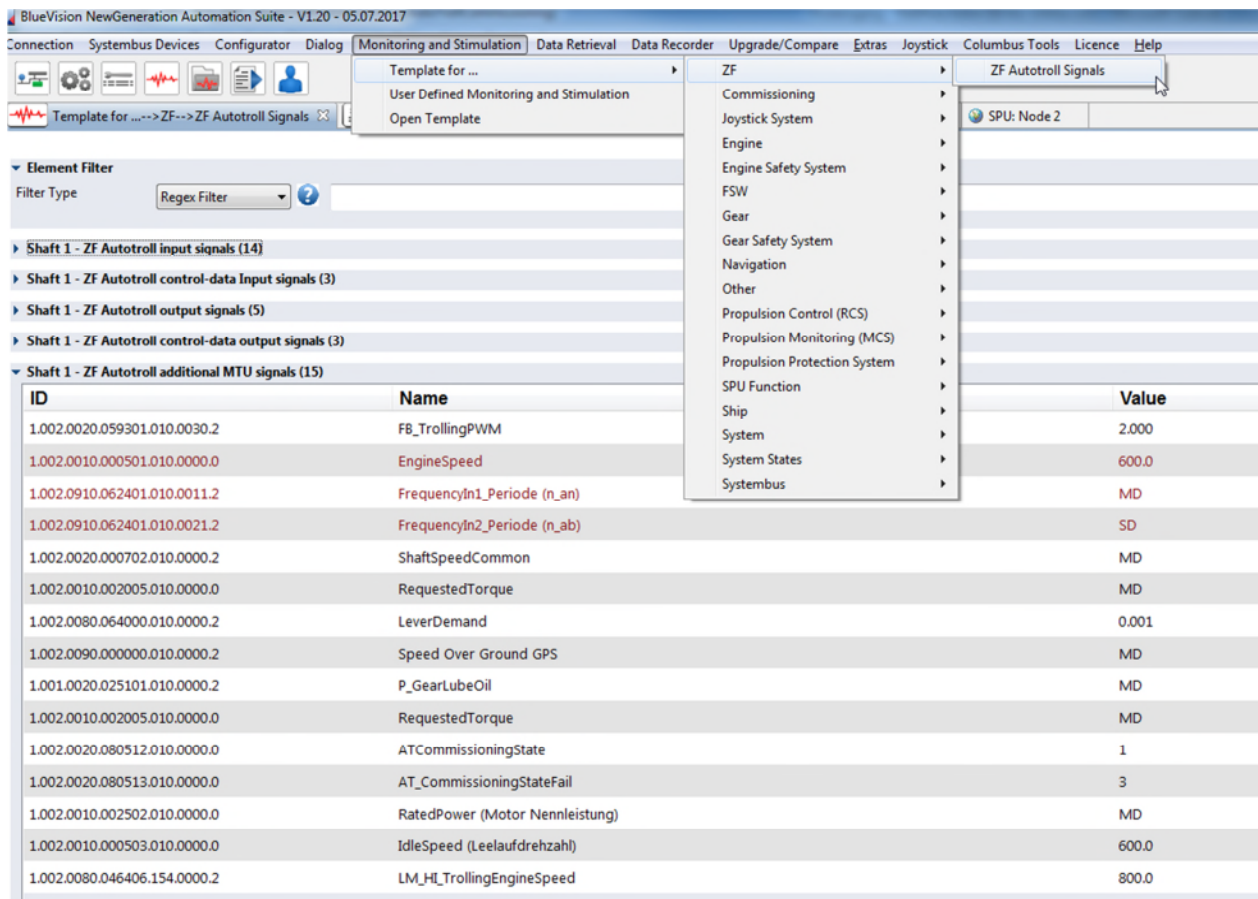
4 Diagnosis

All relevant parameters for Autotroll are available via the "ZF Parameters" template.

Dialog:



Monitoring:



4.1 Monitoring during automatic commissioning procedure

During this time, various actions are performed in the background, which can be monitored via the status variable **ATCommissioningState** (0020.080512.010.000.0).

These actions are divided into 5 steps and are represented in the status variable with the values 1-5. Thus it can be seen in which step the automatic commissioning procedure is currently located.

The corresponding step is therefore equivalent to the displayed value of the status variable:

Value	Meaning
1	The above conditions are being checked but are not met. If they are fulfilled, proceed to step 2.
2	Conditions are fulfilled and measured values are collected and averaged. If the standard deviation is too high (for example, acceleration or too high jitter), it will be stopped and started again with step 1. If OK, then go to step 3.
3	Number of teeth drive side (engine speed sensor) is calculated and stored. If unsuccessful, then step 1 is started again. If successful, then go to step 4.
4	pZF_AT_CommissioningDone is set to True. If unsuccessful, then step 1 is started again. If successful, then go to step 5.
5	Prozedur erfolgreich durchlaufen. Trolling ist nun freigeschaltet und kann aktiviert werden. Procedure was completed successfully. Trolling is now unlocked and can be activated. Nevertheless, the following parameters are monitored for change (should the value of one of the parameters change, step 1 is restarted): GearTransmissionRatio (0020.080302.012.000.1) pRotationSpeed2_TeethCount (0910.062402.012.021.3) pZF_AT_CommissioningDone (0020.080441.012.000.1)

If the automatic commissioning procedure can't be completed successfully, the status variable **AT_CommissioningStateFail** (0020.080513.010.000.0) shows which error status it is.

Value	Meaning
0	no errors
1	basic conditions not fulfilled
2	IO-Manager FI2 mode not activated or wrong configured
3	measurement not stable or not valid
4	invalid teethcount
5	Not possible to write calculated teeth-count drive side in the DOC (DataObjectContainer in SPU)
99	AT Commissioning disabled (pDisable_ZF_AutoCommissioning)

Whether the automatic procedure was successfully completed can be recognized (without activating trolling) by the fact that the parameter **pZF_AT_CommissioningDone** has the value "True" and the status variable **ATCommissioningState** has the value 5.

The automatic commissioning procedure can be triggered again by changing the value of the parameter **pZF_AT_CommissioningDone** from "True" to "False".

Note:

By deactivating and reactivating the trolling type "ZF Autotroll Inside" in the Configurator, the automatic commissioning procedure is not triggered again.

4.2 Check shaft speed

Is there ever a sensible shaft speed displayed?

Based on the known transmission ratio of the gearbox, the shaft speed can be calculated.

The calculated speed and the measured in the engaged state speed (RotationSpeed1 and ShaftSpeedCommon) should not differ much.

The RCS calculates the shaft speed based on the transmission ratio also for other purposes:

ShaftSpeedCalculated (0020.000702.010.010.2)

If the two speeds differ, the set number of teeth (respectively the corresponding entry in the configurator)

pRotationSpeed1_TeethCount (0910.062402.012.011.3) does not match with the reality.

SysID for the frequency/period duration measurement:

FrequencyIn1_Periode (0910.062401.010.011.2)

SPU-channel: FI1MR

pFrequencyIn1_Mode (0910.062401.012.011.3)

Mode must be set to **501** in MCS and RCS!

SysID for the rotational speed:

RotationSpeed1 (0910.062402.010.010.2)

ShaftSpeedCommon (0020.000702.010.000.2) (indication on the displays)

4.3 Check engine speed

After calculating the parameters (automatic commissioning procedure completed), the measured engine speed (**RotationSpeed2**) must correspond to the engine speed supplied by the ECU (**EngineSpeed**).

SysID for the frequency/period duration measurement:

FrequencyIn2_Periode (0910.062401.010.021.2)

SPU-channel: FI2MR

pFrequencyIn2_Mode (0910.062401.012.021.3)

Mode must be set to **501** in RCS and **0** in MCS!

SysID for the rotational speed:

RotationSpeed2 (0910.062402.010.020.2)

4.4 Error handling / alarming

If the automatic commissioning procedure has been completed successfully, but when the trolling function is activated, the "**AL Trolling System Failure**" alarm is triggered, the following SysIDs (ZF messages) can be helpful for the diagnosis:

SysID **Autotroll_Trans_Warn_Err** (0020.080502.010.000.0)

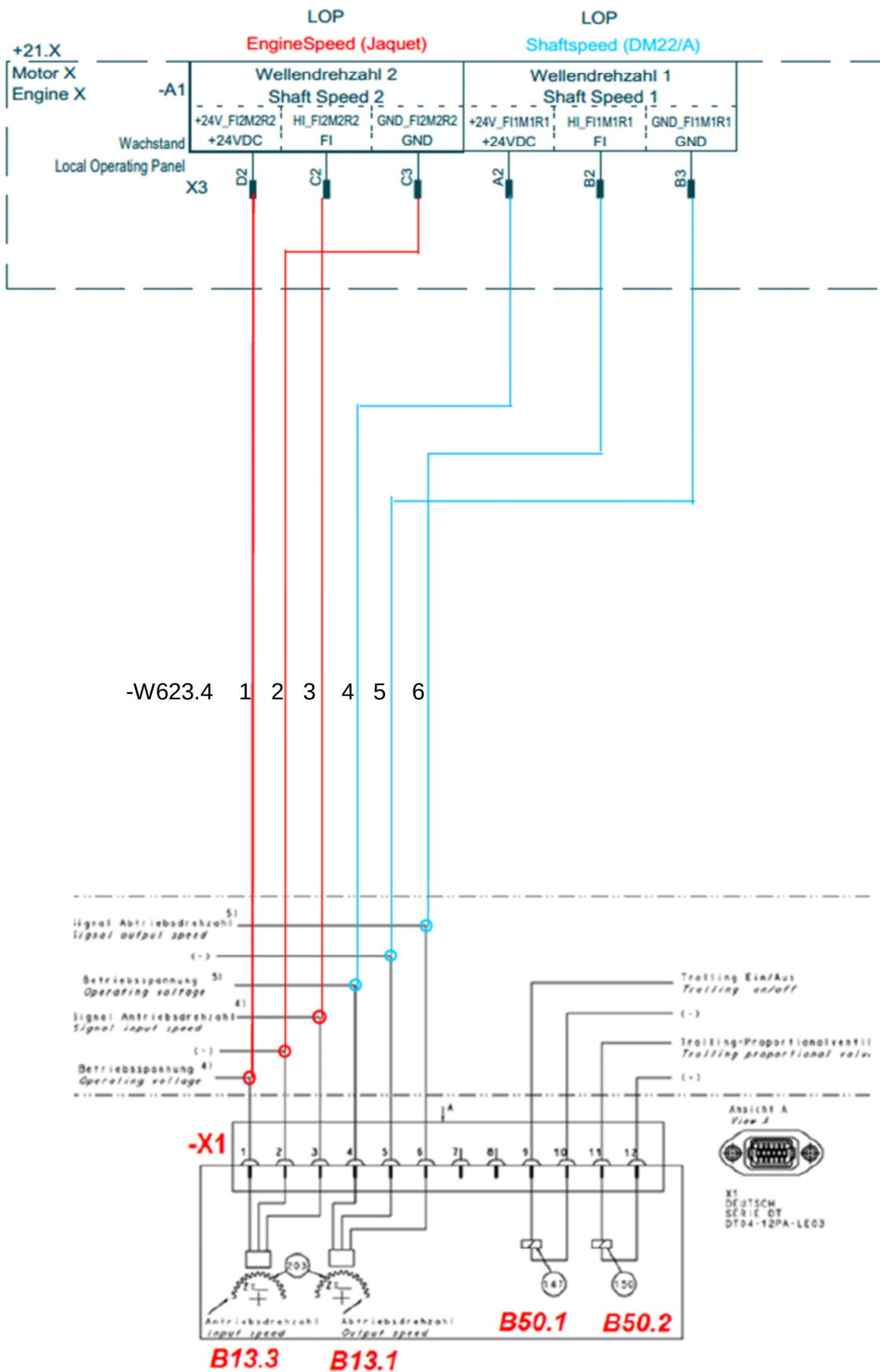
Wert	Bedeutung
0	ZF_NO_ERROR
1	ZF_FAIL_POINTER
2	ZF_FAIL_IF_VERSION
10	ZF_FAIL_N_AN
11	ZF_FAIL_N_AB_DEF
12	ZF_FAIL_N_AB_RED
20	ZF_FAIL_VALVE_ENABLE
21	ZF_FAIL_VALVE_TROLL
30	ZF_FAIL_PLAUSI_N_AB
31	ZF_FAIL_PLAUSI_PARA_MISMATCH
0xFFFFFFFF	ZF_FAIL

An error 30/31 indicates, among other things, that the engine and shaft speeds do not match the set ratio.

SysID **Autotroll_Trolling_Status** (0020.080501.020.000.0)

Wert	Bedeutung
0	ZF_TROLL_OFF → Trolling not activ
1	ZF_TROLL_WARN_OFF → Trolling activ, no warnings
2	ZF_TROLL_WARN_SPEED → Trolling activ, engine speed warning (too high)
3	-
4	ZF_TROLL_SPEED_HIGH → Trolling preselected, but clutch closed because of too high engine speed
5	ZF_TROLL_NOT_ADJUSTABLE

5 Sensor wiring (system cable -W623.4)



6 Number of teeth (impulses per rotation) output shaft (Shaft speed)

GETRIEBETYP GEARBOX TYPE	BEDINGUNG/ REQUIREMENT	IMPULSE PRO UMDREHUNG ABTRIEBSWELLE IMPULSES PER ROTATION OUTPUT SHAFT
4) ZF 2000; ZF 2000 A; ZF 2000 V; ZF 2050; ZF 2050 A; ZF 2050 V ZF 2060; ZF 2060 A; ZF 2060 V; ZF 2070; ZF 2070 A; ZF 2070 V ZF 2075; ZF 2075 A; ZF 2075 V; ZF 2150; ZF 2150 A; ZF 2150 V		62
ZF 2200; ZF 2250; ZF 2260; ZF 2270; ZF 2300; ZF 2350; ZF 2360; ZF 2370 ZF 3000; ZF 3000 A; ZF 3000 V; ZF 3050; ZF 3050 A; ZF 3050 V ZF 3055; ZF 3055 A; ZF 3055 V; ZF 3060; ZF 3060 A; ZF 3060 V ZF 3070; ZF 3070 A; ZF 3070 V		77
ZF 3310; ZF 3311; ZF 3350; ZF 3351; ZF 3355; ZF 3356 ZF 3360; ZF 3361; ZF 3710; ZF 3711; ZF 3751		97
ZF 4660; ZF 4661; ZF W4610; ZF 7540; ZF 7541; ZF 7561; ZF 7566; ZF 7640; ZF 7641 ZF 7645; ZF 7646; ZF 7661; ZF 7666; ZF W7500; ZF W7540; ZF W7610; ZF W7615; ZF W7645		116
ZF 4540; ZF 4540 A; ZF 4540 V; ZF 4600; ZF 4600 A; ZF 4600 V; ZF 4640 ZF 4650; ZF 4650 A; ZF 4650 V	n < 756xi 2) ODER / OR n ≥ 1080xi	77
	756xi ≤ n < 1080xi	87
ZF 5000; ZF 5000 A; ZF 5000 V; ZF 5050; ZF 5050 A; ZF 5050 V ZF 5055; ZF 5055 A; ZF 5055 V		90
ZF 5300; ZF 5350; ZF 5355; ZF 5360		102
ZF 7500; ZF 7549; ZF 7550; ZF 7550 A; ZF 7550 V ZF 7550 NR2; ZF 7550 NR2H; ZF 7550 NR2B ZF 7600; ZF 7600 A; ZF 7600 V ZF 7600 NR2; ZF 7600 NR2H; ZF 7600 NR2B ZF 7650; ZF 7650 A; ZF 7650 V ZF 7650 NR2; ZF 7650 NR2H; ZF 7650 NR2B	n < 641xi 2) ODER / OR n ≥ 936xi	90
	641xi ≤ n < 936xi	100
ZF 7500 C		3)
ZF 9000; ZF 9000 A; ZF 9000 V; ZF 9000 NR2; ZF 9000 NR2H; ZF 9000 NR2B ZF 9050; ZF 9050 A; ZF 9050 V; ZF 9050 NR2; ZF 9050 NR2H; ZF 9050 NR2B ZF 9055; ZF 9055 A; ZF 9055 V; ZF 9055 NR2; ZF 9055 NR2H; ZF 9055 NR2B		114
ZF 9300; ZF 9311; ZF 9350; ZF 9351; ZF 9355; ZF 9356		140